Antranig Baghdassarian

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Education

Cornell University, Ithaca, NY – Bachelor of Science in Biomedical Engineering — Expected May 2027

- Emphasis in Electrical Engineering | Certifications in LEAN Yellow Belt & Engineering Leadership — GPA: 3.62

Research Experience

Student Researcher, Dr. Edwin Kan's Lab – Ithaca, NY — Jan 2026 – Present

- Assist with experimental design for non-invasive blood-flow monitoring through the brain using a 900 MHz sensing system.
- Evaluated Doppler-based blood-flow velocity estimation using a 900 MHz FMCW radar system within GNU Radio.

Undergraduate Researcher (REU), Tamerler Lab, University of Kansas — May – Jul 2024

- Led eight independent bacterial-culture experiments using antimicrobial and AlamarBlue assays to evaluate a novel bifunctional peptide for dental adhesives.
- Analyzed cell viability using Cytation software and collagen fibril formation using atomic force microscopy.
- Presented results of bi-functional peptide to maintain antimicrobial and collagen forming properties at symposium

Student Researcher, Pancreatic Cancer Research Internship — Sep 2022 – Feb 2023

- Analyzed ten pancreatic-cancer patient MRI scans under an NYU professor to investigate more efficient detection of risk factors.
- Used FireVoxel to measure radiologist-marked regions of interest, including cross-sectional area, length, and volume; performed regression analysis to investigate trends.

Work Experience

Mechatronics Engineering Intern, LEADOPTIK - San Jose, CA — May 2026 – Aug 2026

- Developed a functional test rig for the main printed circuit board (PCB) to streamline board-level testing during manufacturing.
- Designed and 3D-printed a catheter evaluation test fixture to streamline setup and improve throughput and repeatability.
- Created a Python script and GUI to automatically email vendors out of compliance, saving 10+ hours.

Quality & Compliance Intern, Johnson & Johnson Vision – Jacksonville, FL — May 2025 – Aug 2025

- Created and redesigned a Tableau inventory and shipment dashboard with automatic quantity updates, saving 5+ hours/month and increasing visibility of stock and shipments.
- Prototyped an automated product crate validation process with Power Automate via GxP tools, decreasing occurrence of human error & projected savings of 9+ hours/month

Technical and Entrepreneurial Experience

Technical Project Manager, Empathera – Ithaca, NY Mar 2024 – Present

- Prototyped an electrical muscle stimulation device to help prevent pressure ulcers and reduce caregiver strain.
- Won 5 awards in various business competitions, leading to +$10k in non-dilutive funding
- Featured in various articles such as: Weill Medicine, Cornell Chronicle, and Global Cornell
- Developed the stimulation method and device algorithm following the hackathon; organized team deliverables and meetings.
- Consulted electrical, biomedical, and materials-engineering advisers and entrepreneurship mentors for product feedback and guidance.

Electrical Engineering Team Co-Lead, Nexus Project Team – Ithaca, NY Feb 2024 – Jan 2026

- Managed weekly subteam meetings, member deliverables, cross-team communication, the subteam budget, and a $100k+ Altium PCB-design sponsorship.
- Led electrical integration of seven devices onto a rover for removing microplastics from beaches; trained four new members using Arduino starter kits.
- Selected pogo-pin contacts and receptacles, iterated charging-contact CAD in Fusion 360, and soldered prototypes for an autonomous rover charging interface.
- Selected a base-station battery and DC–DC charge controller; investigated MPPT connection order and relay sequencing for solar-to-rover charging.
- Contributed to Raspberry Pi motor/encoder integration through Python and GPIO work, encoder measurements, PWM and wheel-speed analysis, and code review.
- Presented charging-system and contact-design progress in the Fall 2025 CDR, documenting prototype iterations and planned integration tests.
- [MORE INFO NEEDED] Confirm co-lead dates versus overall team membership, personal ownership of final integration, and measured charging/contact performance.

Leadership Experience

Events & Planning Chair, Cornell Maker Club – Ithaca, NY Feb 2024 – Present

- Organize 20+ workshops (30+ people), annual makeathon, while utilizing event feedback and brainstorming new events

Leadership Roles, Biomedical Engineering Society – Ithaca, NY Oct 2023 – Present

- Establish club's website, official university email, society membership & monitor emails, increasing visibility of club
- Lead meetings and take minutes, brainstorm and organize events while establishing interclub connections for longevity
- Serve as president; previously served as freshman representative, secretary, and vice president of the Biomedical Engineering Society.
- Lead the planning and execution of academic, professional, and social events for 200+ members, coordinating with student, faculty, and campus organizations.
- Develop programming initiatives to strengthen student-faculty connections, expand professional development opportunities, and increase member engagement

Peer Advisor, Cornell Engineering Advising – Ithaca, NY Sep 2024 – December 2025

- Outstanding Peer Advisor Award recipient, given to 2 out of 83 advisors for initiative in planning lessons and positivity
- Advised 25+ students/year on College of Engineering resources, opportunities, requirements, and questions about the transition to college.

Mastermind Circle Facilitator, Cornell Engineering Leadership Spring 2025

- Helped develop a 10-week program across four circles serving 17 participants; 11 earned completion certificates.
- Facilitated weekly Mastermind Circle meetings focused on participants' personal, academic, and research goals.
- Guided exercises in SMART goals, core values, question storming, self-feedback, self-saboteur awareness, and charting commitments.
- Co-prepared weekly presentations and exercises; used facilitator debriefs to improve session flow and address group challenges.
- Collaborated on task ownership, time-boxed agendas, and workload distribution to match team members' capacity.

Business Team CLead, XR Club Feb 2024 – Oct 2024

- Connected and organized a collaboration between the Johnson Museum and XR Club for a $50k+ exhibit, helping ensure the virtual-reality component ran smoothly.

Corporate Subteam Member, IEEE Started Sep 2023

- Developed relationships with companies and students through sponsored technical talks and résumé reviews.
- Invited PhD, master's, and undergraduate students to a Marvell technical talk and résumé review attended by 40 students.

Founder and President, Great Neck South High School Esports Club Sep 2022 – Jun 2023

- Established a High School Esports League chapter, connecting students with national competitions offering scholarship opportunities.
- Organized four game nights and schoolwide intergrade competitions in collaboration with student council.

Vice President, Astronomy Club Sep 2021 – Jun 2023

- Helped brainstorm activities, set meeting reminders, and take meeting notes; raised $270 through bake sales.
- Co-scheduled open-observatory nights and announced events through the school PA system.

Projects

Bluetooth Low Energy (BLE) Macropad June 2026 – Present

- Design small 4-layer PCB via KiCad, functioning as Bluetooth keyboard (4 keys, macropad) mapped to media keys
- Develop firmware for nRF52840 via Zephyr and BLE over GATT, implemented support for multi-host profile switching

Dye-Sensitized Solar Cell Workshop (2) Aug 2022, Aug 2023

- Designed and led virtual and in-person workshops for 20+ rural Armenian students on building blackberry dye–based solar cells and understanding solar technology, energy policy, and Armenia's solar development plan; partnered with MAQOOR and featured in Mirror Spectator
- Taught students to build solar cells using blackberry dye, titanium dioxide semiconductor paste, graphite, iodine, and conductive glass.
- Covered solar-panel operation, benefits and limitations, and Armenia's electricity outlook and solar-development plans.

Book Publication

Author, Close Encounter with This Weird Robot Thing Feb 2017

- Wrote and published a 40-page children's book about a child-and-robot duo saving the world; met weekly with the publisher to discuss publication logistics.
- Featured in the Great Neck Record.

Selected Honors and Awards

- Cornell Health Hackathon Grand Prize — Empathera.
- Outstanding Peer Advisor Award — selected as one of two recipients among 83 Cornell Engineering peer advisers.
- Dean's Award — Great Neck South High School, 2023.
- AP Scholar Award — 2022.
- DECA State Awards — Top 10 Test in Integrated Marketing Campaign: Event and Top 10 Overall, 2022.

Skills & Interests

Software: Fusion 360, SOLIDWORKS, KiCad, Ansys Fluent, Power Automate, Tableau, Bambu Studio, LabVIEW
Additional tools: Altium Designer, FireVoxel, GNU Radio, Cytation, PyAnsys, Jira, DaVinci Resolve; Arduino and ESP32.
Certifications: LEAN Yellow Belt, Leadership in Engineering Field
Languages: Python, Java, MATLAB, C, R, Spanish | **Development:** Soldering, Programming with Microcontrollers
Interests: Electronics**,** Engineering How-To Videos, Running, Volleyball, My Hero Academia, and Kaizen Mentality